Human Computer Interaction

Evaluation Techniques

Evaluation

- Evaluate design with expert
  - Cognitive Walkthrough
  - Heuristic Evaluation

- Evaluate design with people
  - User testing

Evaluate design with expert

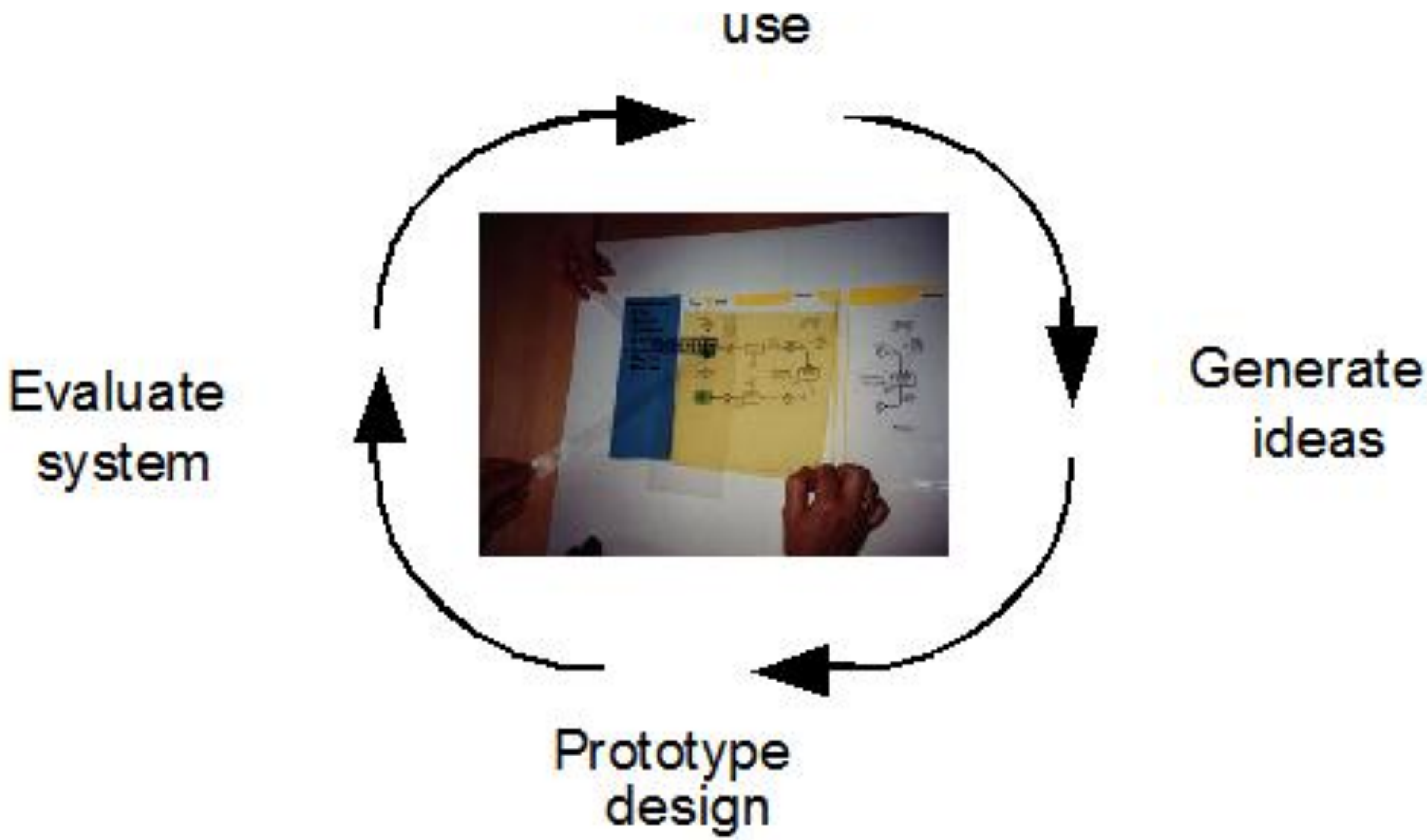
- An interface is evaluated by usability experts rather than end users.

- Usually several evaluators work independently, then debrief together later.

When to get design critique from experts?

- Before user testing.
  - Don’t waste users on the small stuff. Critique can identify minor issues that can be resolved before testing, allowing users to focus on the big issues.
- Before redesigning.
  - Don’t throw out the baby with the bathwater. Critique can help you learn what works and what should change.
- When you know there are problems, but you need evidence.
  - Perhaps you've received complaints from customers or found yourself stumbling around your own site. Critique can help you articulate problems and provide you with ammunition for redesign.
- Before release.
  - Smooth off the rough edges.

System Evaluation



Evaluate design with expert

Cognitive Walkthrough

Cognitive Walkthrough: Introduction

- Proposed by Polson
  - evaluates design on how well it supports user in learning task
  - usually performed by expert in cognitive psychology
  - expert ‘walks through’ design to identify potential problems using psychological principles
  - main focus “how easy a system is to learn”

Cognitive Walkthrough: Introduction

- Focuses on evaluating a design for ease of learning
- Experts evaluate a proposed interface in the context of one or more specific user tasks
- Experts walk through the tasks. At each step, ask yourself four questions.
    1. What is the user trying to achieve at this point?
    2. Will users see the control (button, menu, switch, etc.) for the action? (**Visibility**)
    3. Once users find the control, will they recognize that it produces the effect they want? (**Mapping**)
    4. After the action, will users understand the feedback they get so they can go on to the next action with confidence? (**Feedback**)

Cognitive Walkthrough: Procedure

- Define the inputs
- Gather the walkthrough team
- Walk through the action sequences for each task
- Record critical information
- Revise the interface to fix the problems

Performing the Cognitive Walkthrough

- Define the inputs
  - Identify users and tasks
  - Create a description (screenshots, storyboard) or implementation (rapid prototype) of the interface
  - Define the action sequences for completing each task
- Gather the team
  - Facilitator maintains the pace of the discussion

Performing the Walkthrough, cont.

- Walk through the action sequences for each task
  - Participants walk through (discuss) the tasks with respect to the interface (prototypes or screenshots) and action sequences
  - They try to tell a credible story
    - What is the user trying to achieve at this point? What is the user's goal and why is it their goal?
    - What actions are obviously available in the interface?
    - Does the label for the correct action match the user's goal?
    - If the user performs the correct action, will they get good feedback?

Performing the Walkthrough, cont.

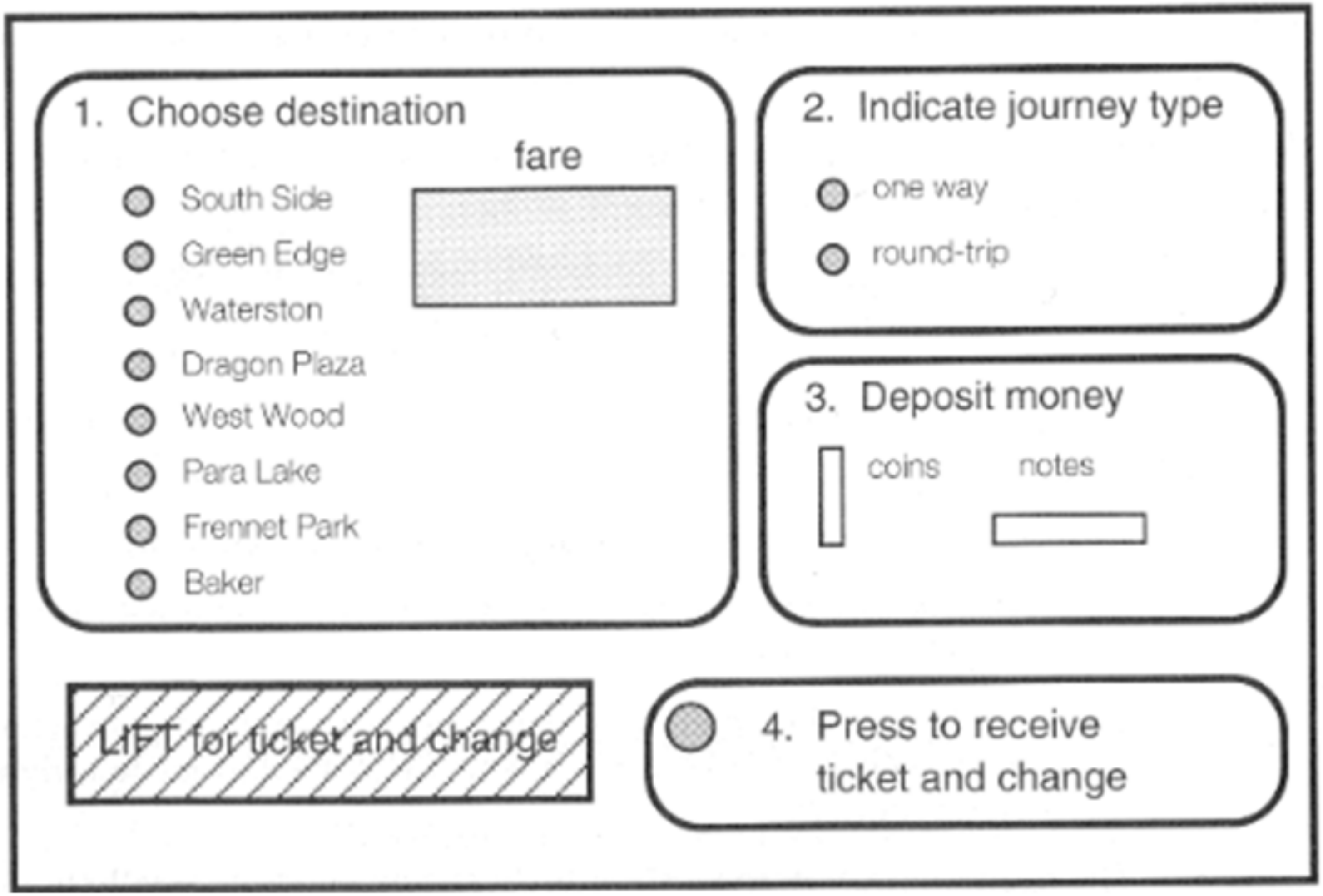
- Record critical information
  - The credible success (or failure) story
  - Assumptions (about tasks and user's skill)
  - Problems (and suggested solutions)
- Revise the interface to fix the problems
  - Re-implement rapid prototype or create new screenshots

Cognitive Walkthrough Example - Rapid Transit Ticket Machine

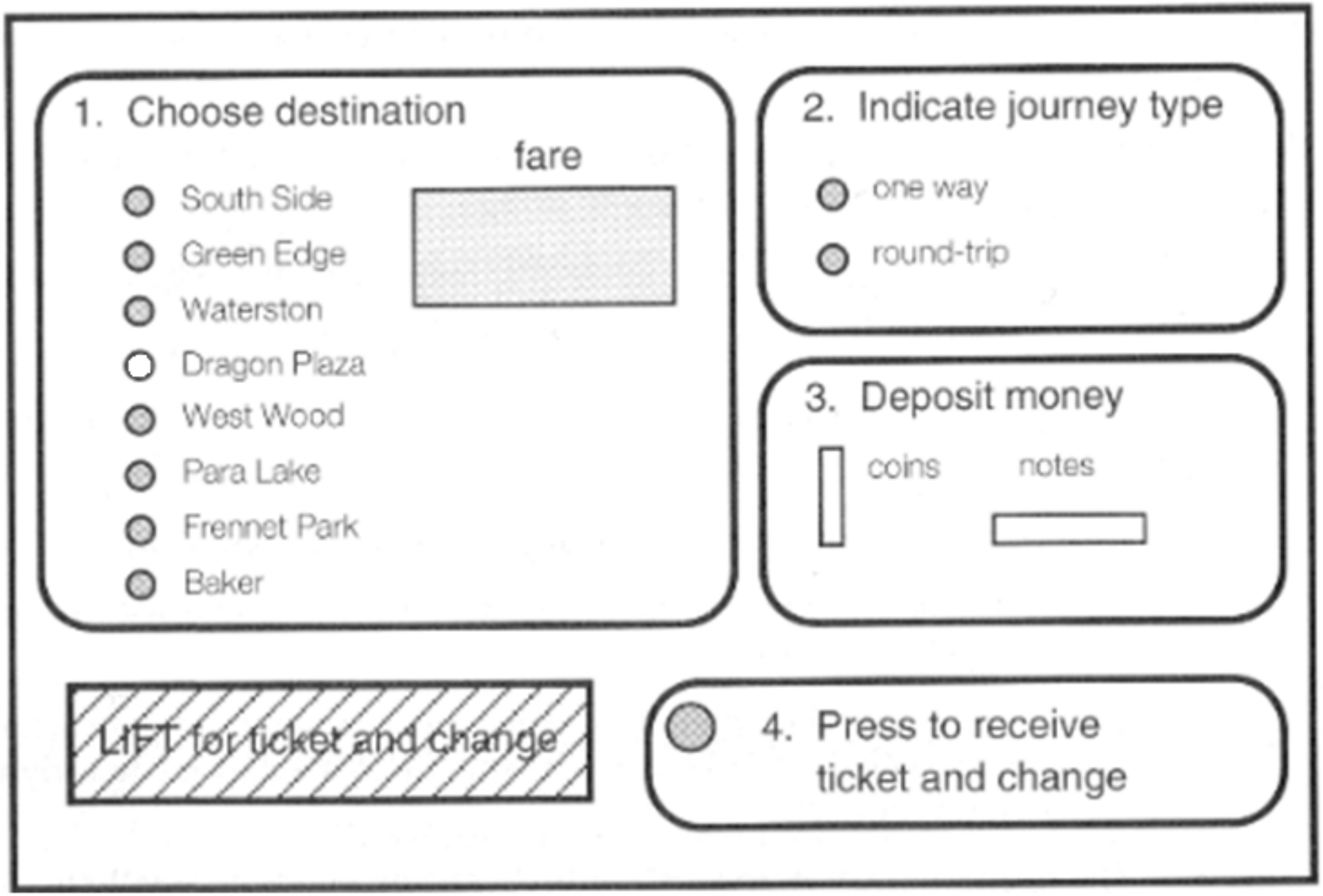
- User wishes to purchase a round-trip ticket to Dragon Plaza.
- The ticket costs $17.50 but the user doesn't know this yet.
- The user has only $10, but doesn't know this yet either.

This example is taken from http://asi-www.informatik.uni-hamburg.de/informatik/SE_Evaluation/html/pages/walk.htm

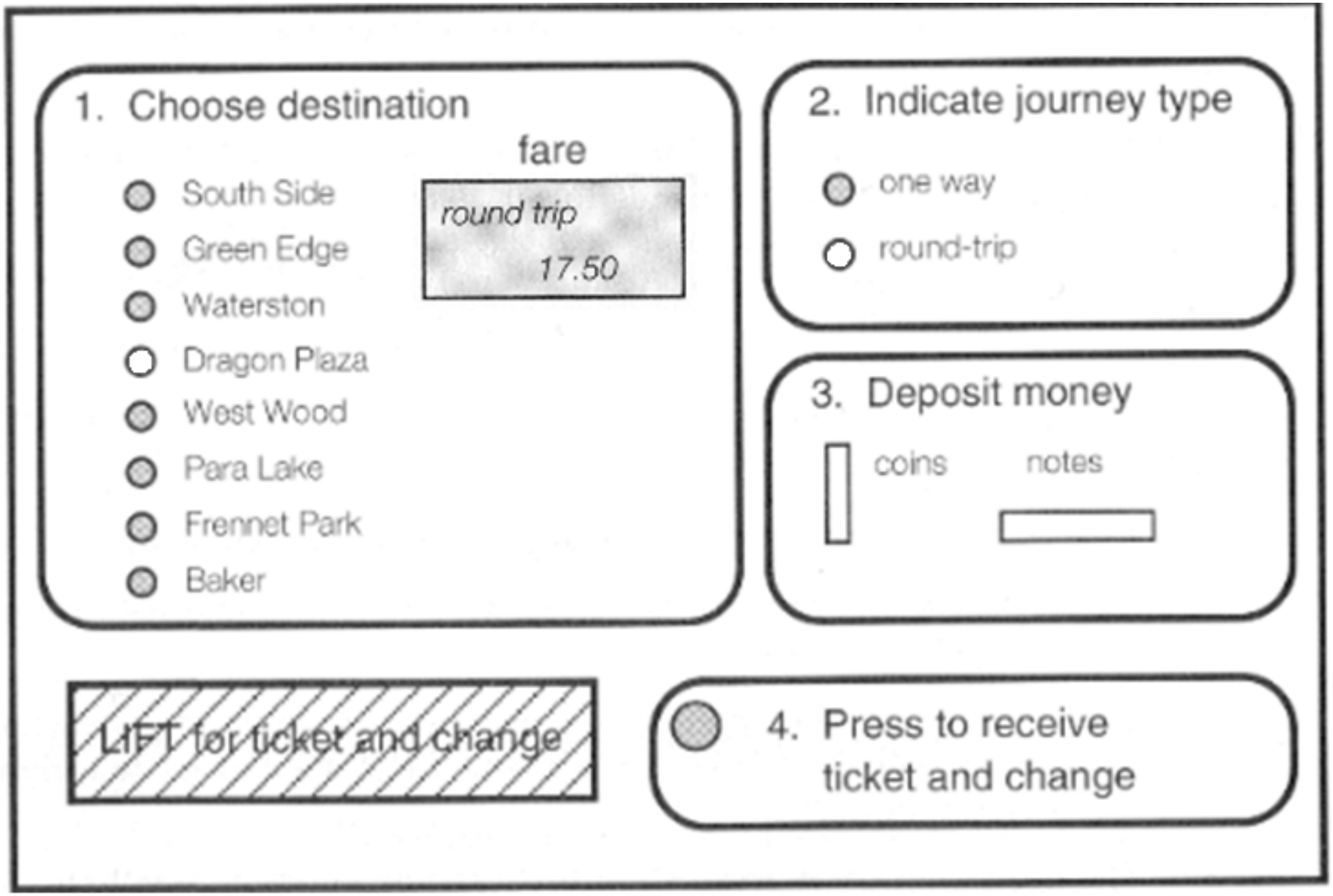
Step 1: Enter Destination



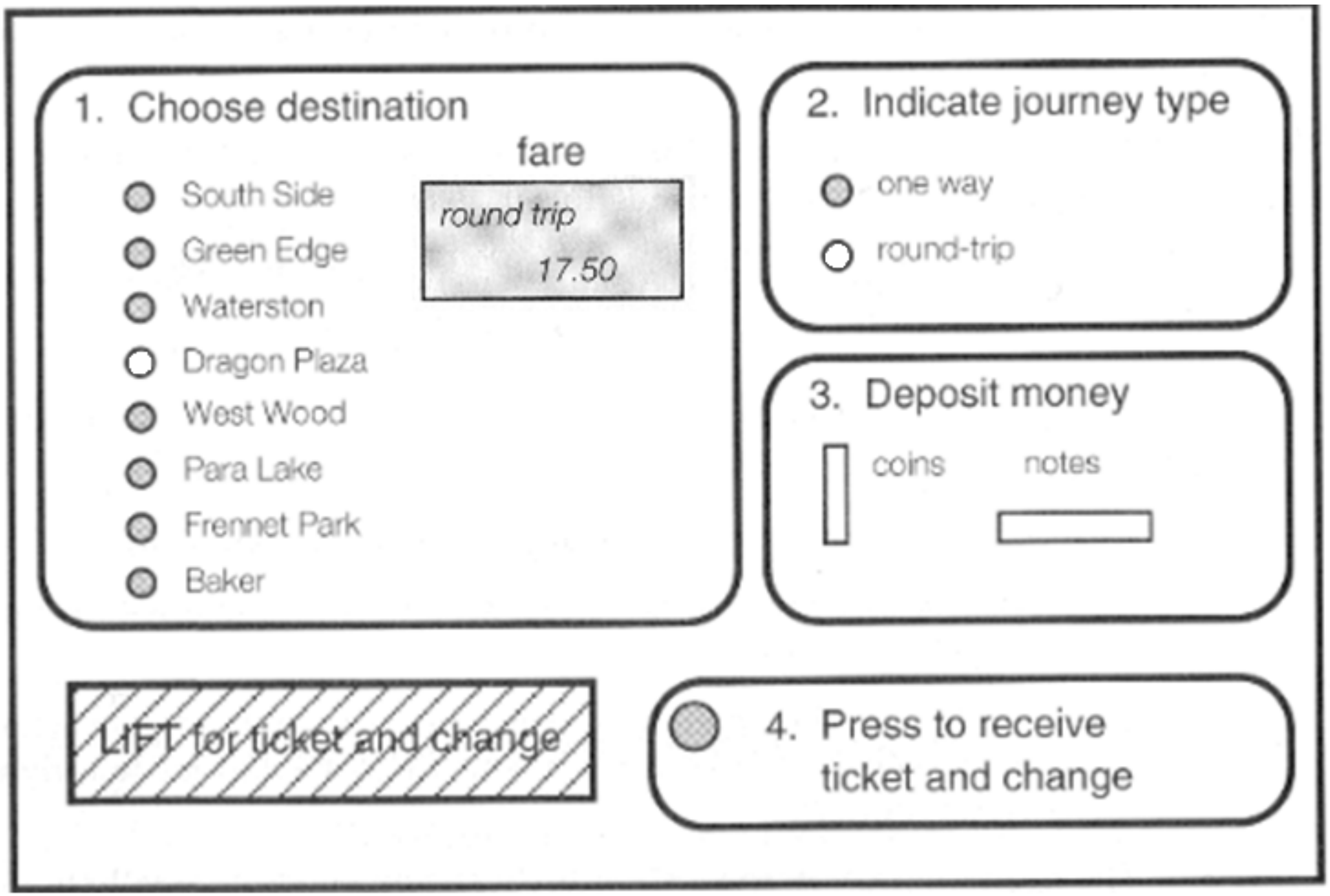
- Design Flaw no. 1: Option to indicate journey type first is not made sufficiently evident.



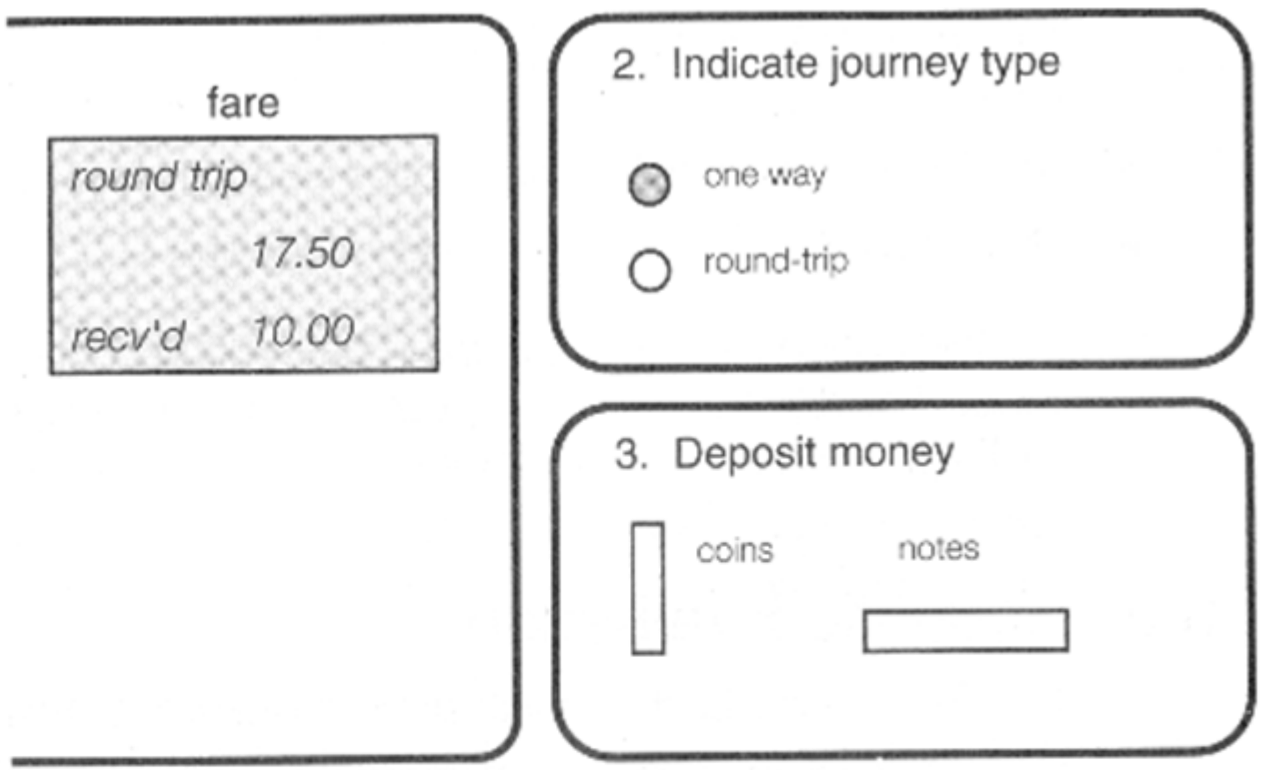
Step 2: Enter Journey Type



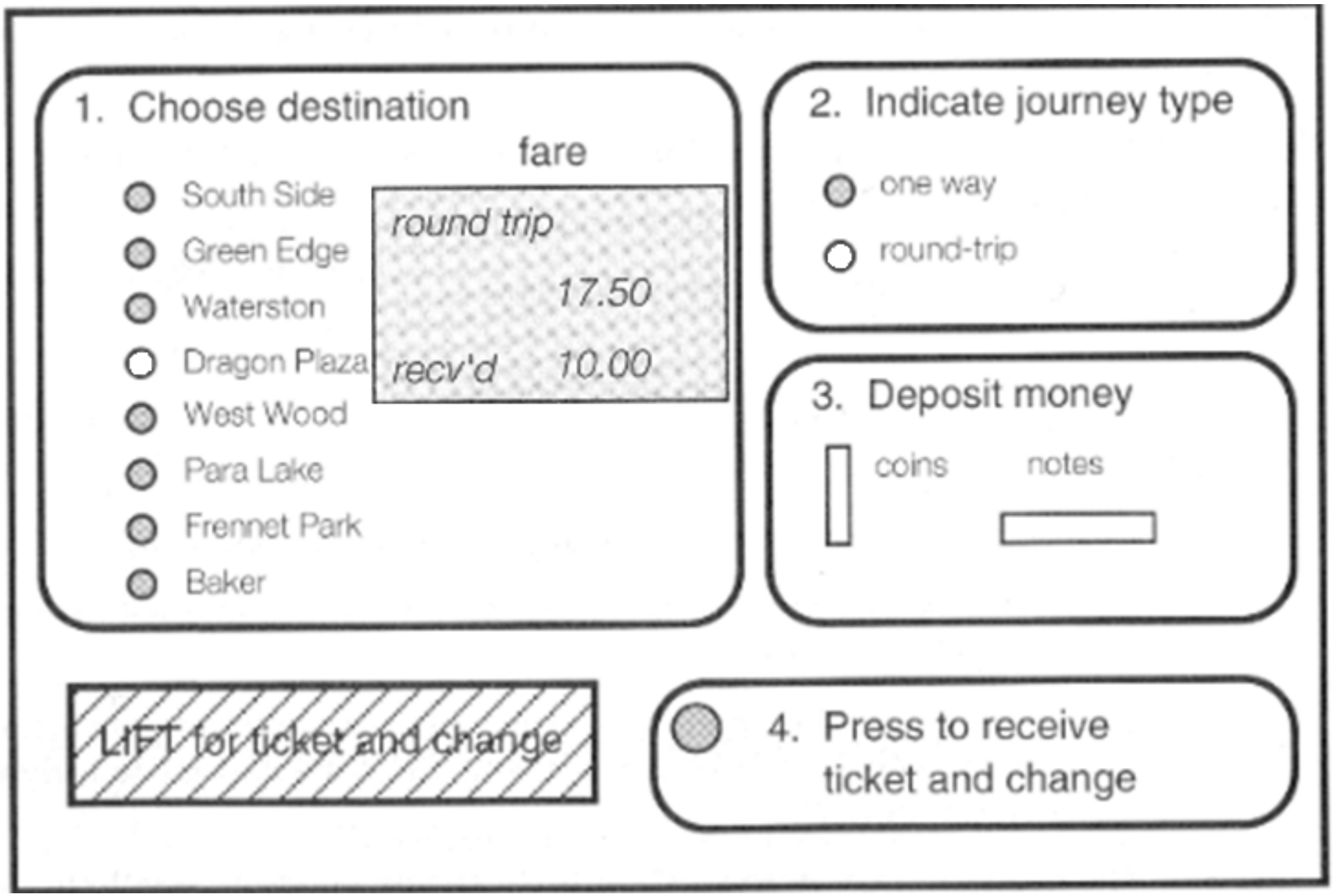
Step 3: Deposit money



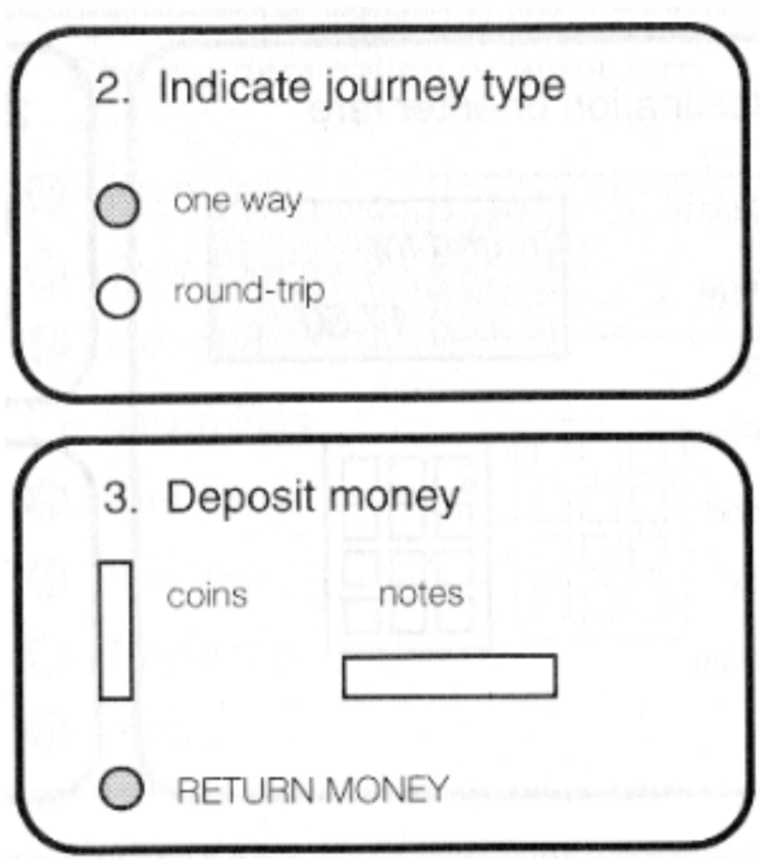
- Design Flaw no. 2: No display of total money received
- Solution:



Step 4: Retrieve $10 from machine since it wasn't enough



- **Design Flaw no. 3: No means of retrieving money deposited.**
- **Solution:**



Cognitive Walkthrough

Another Example

Inverting a portion of an image (Example)

- Users
  - We want novice users of Photoshop to be able to invert selections of an image with little or no training;
  - Assume that user's have had experience with other imaging programs
- Tasks
  - Select a subregion of an image and invert it

Example, cont.

- Action Sequences
  - Zoom display to area of interest
  - Select the Lasso Tool
  - Select the subregion of the image
  - Select Inverse from the Image menu

Photoshop Interface

Description of Interface

- Photoshop presents
  - a toolbar (far left)
    - vertically arranged
    - Assume that novice users are unfamiliar with the toolbar's icons
  - the image (center)
  - a control panel (far right)
    - Assume that novice users are unfamiliar with the operation and purpose of the control panel

Zoom in on Face

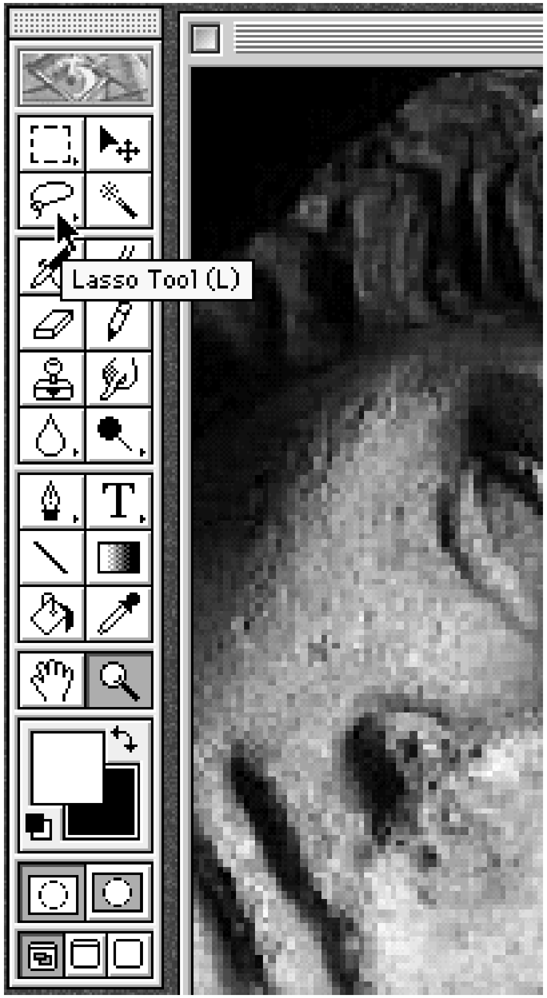
Action: Zoom in on Image

- What's the user's goal, and why?
  - The user wants to specify the portion of the image to invert exactly. Zooming in on the region of interest helps to increase the accuracy of the selection
- Is the action obviously available?
  - The default tool in Photoshop is the Zoom tool; if the user has just started Photoshop its the current tool
  - Novice users may have to search for this tool on the toolbar if they need it later on
  - This tool uses the magnifying glass as its icon

Zoom in on Image, cont.

- Does the action or label match the goal?
  - No label involved here, however magnifying glass icon represents task well
  - Clicking on image, zooms the tool
  - Dragging on image, specifies zoom region more accurately
  - Novice users may click rather than drag
- Is there good feedback?
  - Yes, Photoshop instantly zooms the image

Select Lasso Tool



Select the Lasso Tool

- What's the user's goal, and why?
  - They need a tool to select a portion of the image
- Is the action obviously available?
  - They are familiar with the lasso tool from other image programs
  - The lasso icon is available at the top of the toolbar (increasing the chance that it will be seen)
  - The tooltip provides confirmation of the icon's information

Select the Lasso Tool, cont.

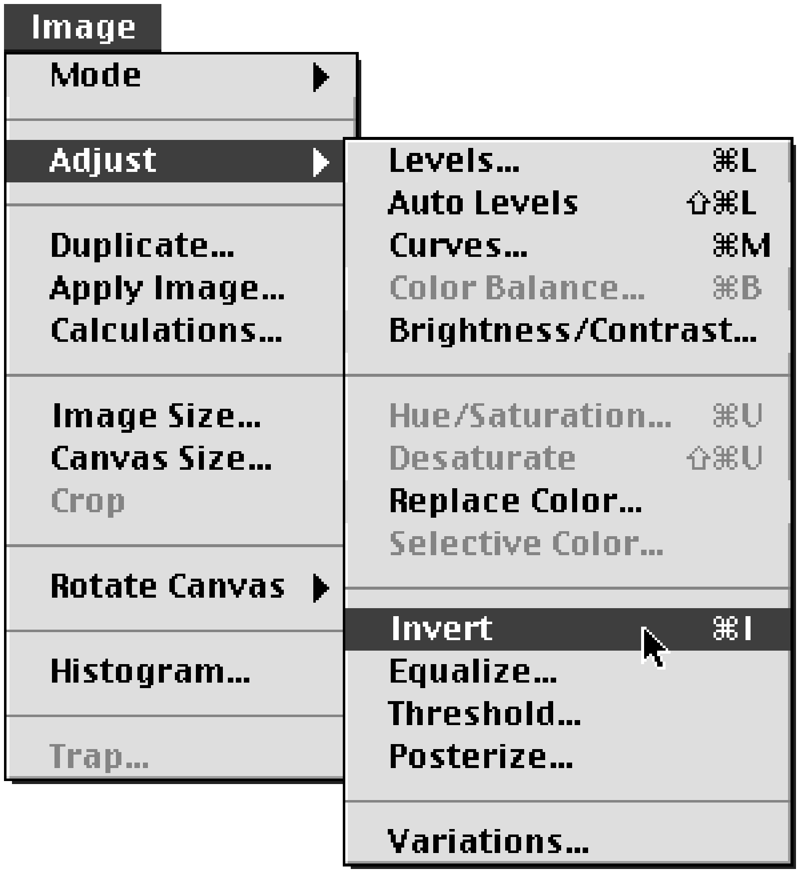
- Does the action or label match the goal?
  - Tooltip serves as label and confirms the meaning of the familiar lasso icon
- Is there good feedback?
  - Not shown but toolbar icon highlights when selected

Select Image

Select the Image

- What's the user's goal, and why?
  - Select a portion of the image
- Is the action obviously available?
  - Assume novice user is familiar with using the lasso tool
- Does the action or label match the goal?
  - Yes, the lasso tool's sole purpose is selecting regions
- Is there good feedback?
  - Yes, the lasso tool produces a "rubber-band" that indicates the current selection

Select Invert Operation



Invert the Image

- What's the user's goal and why?
  - The overall task is to invert a region of the image
- Is the action obviously available?
  - No, previous experience will lead them to look for action in the menus
- Does the action or label match the goal?
  - Yes, but the invert operation is buried in a submenu called Adjust within the Image menu; novice users may look for the command in the "Edit" menu
  - Is there good feedback? Yes, the image inverts

Operation Complete

Example Wrap-up

- Action 1: Zoom In
  - Available as default tool; novice users may have to search for tool on subsequent operations
- Action 2: Select Lasso Tool
  - Lasso Icon is located in prominent place on toolbar; novice users are familiar with this tool
- Action 3: Select Image
  - No problem here
- Action 4: Invert Image
  - Invert command is buried in submenu

Possible Improvements?

- The Invert selection command is a common operation yet it is buried in a submenu
  - "Image -> Adjust" may not be intuitive to a novice user
  - Note: It is assigned an intuitive keyboard short-cut (Ctrl+I) which is good!
- Suggestions
  - Move Invert up one level into the Image menu?
  - Place a command for inverting the selection on one of the toolbars?

Evaluate design with expert

Heuristic Evaluation

Heuristic Evaluation

- Proposed by Nielsen and Molich.
- A heuristic is a guideline or general principle or rule of thumb that can guide a design decision
- Heuristic evaluation perform on design specification so it is useful for evaluating early design
- It can also used on prototypes, storyboards and fully functioning systems, therefore it is flexible and cheap approach
- design examined by experts to see if these are violated

Heuristic Evaluation

- Example heuristics
  - system behaviour is predictable
  - system behaviour is consistent
  - feedback is provided

- Heuristic evaluation `debugs' design.

Heuristic Evaluation

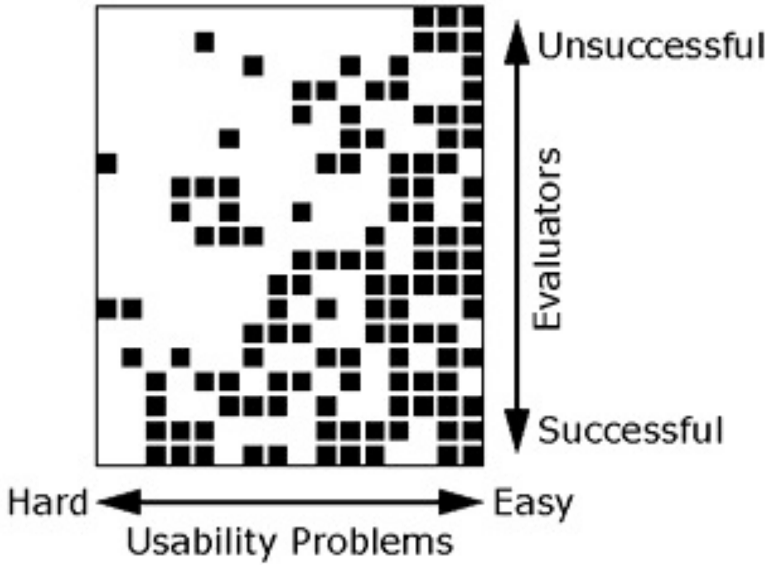
- Helps find usability problems in a design
- Small set (3-5) of evaluators examine UI
  - independently check for compliance with usability principles ("heuristics")
  - different evaluators will find different problems
  - evaluators only communicate afterwards
    - findings are then aggregated
- Can perform on working UI or sketches

Evaluators' Process

- Step through design several times
  - Examine details, flow, and architecture
  - Consult list of usability principles
  - ..and anything else that comes to mind
- Which principles?
  - Nielsen's "heuristics"
  - Category-specific heuristics from e.g., design goals, competitive analysis, existing designs
- Use violations to redesign/fix problems

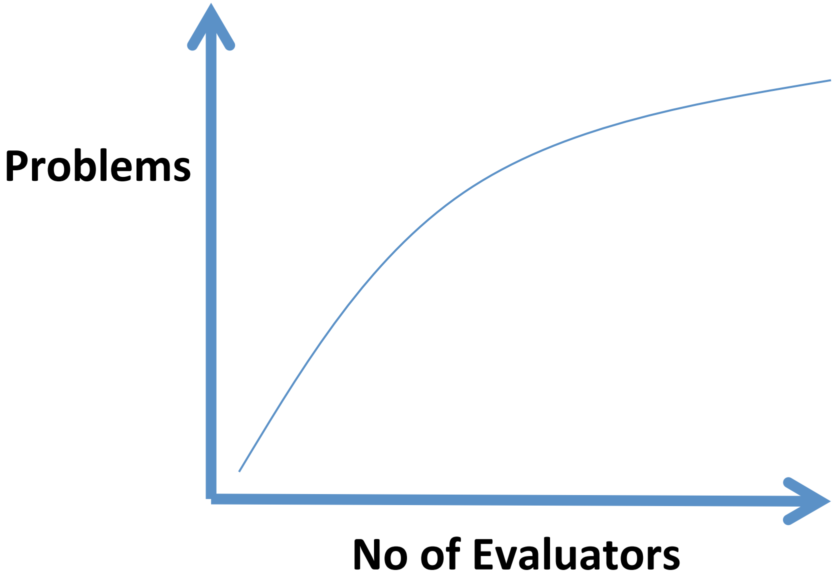
Why Multiple Evaluators?

- No evaluator finds everything
- Some find more than others

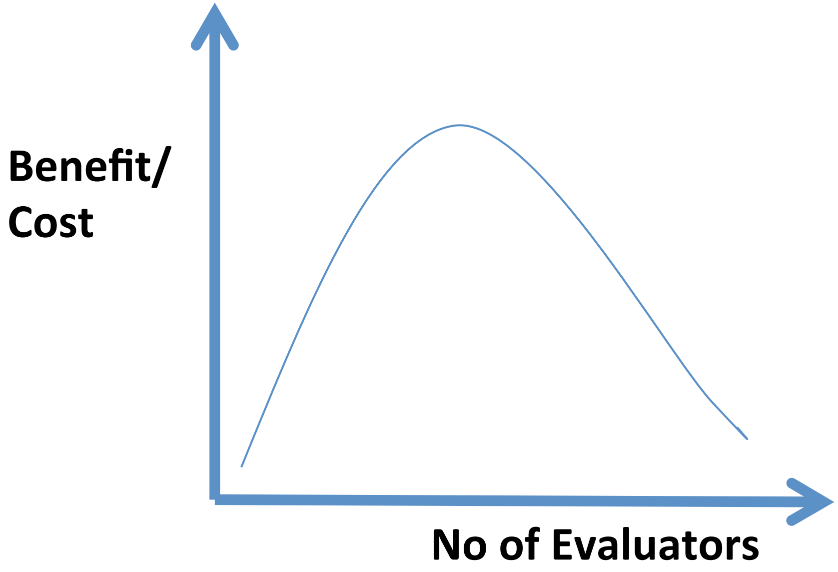


Decreasing Returns

Problems found



Benefits/Cost ratio



Heuristic Evaluation: Cost-effective

- In one case: benefit-cost ratio of 48
  - estimated benefit $500,000; cost $10,500

- Severe problems found more often

- Single evaluator achieves poor results
  - only finds 35% of usability problems
  - 5 evaluators find ~ 75% of problems

Heuristics Evaluation vs. User Testing

- Heuristic Evaluation (H.E.) often faster
  - 1-2 hours each evaluator
- H.E. results come pre-interpreted
- User testing is more accurate
  - takes into account actual users and tasks
  - H.E. may miss problems & find “false positives”
- Valuable to alternate methods
  - find different problems
  - don’t waste participants

Phases of Heuristic Evaluation

1. Pre-evaluation training: give evaluators needed domain knowledge and information on the scenario
2. Evaluation: individuals evaluate and then aggregate results
3. Severity rating: determine how severe each problem is (priority). Can do first individually and then as a group
4. Debriefing: review with design team

How-to: Heuristic Evaluation

- At least two passes for each evaluator
  - first to get feel for flow and scope of system
  - second to focus on specific elements
- If system is walk-up-and-use or evaluators are domain experts, no assistance needed
  - otherwise might supply evaluators with scenarios
- Each evaluator produces list of problems
  - explain why with reference to heuristic or other information
  - be specific and list each problem separately

Severity Ratings

0 - don’t agree that this is a usability problem

1 - cosmetic problem

2 - minor usability problem

3 - major usability problem; important to fix

4 - usability catastrophe; imperative to fix

Severity Ratings Example

Issue: Unable to edit one's weight

Severity: 2

Heuristics violated: User control and freedom

Description: when you open the app for the first time, you have to enter your weight, but you cannot update it. It could be useful if you mistyped your weight, or if one year or two after the first use of the app, your weight has changed.

Debriefing

- Conduct with evaluators, observers, and development team members
- Discuss general characteristics of UI
- Suggest potential improvements to address major usability problems
- Development team rates effort to fix
- Brainstorm solutions

Summary

- **Cognitive Walkthrough**
  - **Who:** new user
  - **What:** performs list of specific tasks
  - **Why:** to evaluate the learnability and to see if the tasks can be performed in the correct sequence of actions they were designed in

- **Heuristic Evaluation**
  - **Who:** system specialist (preferably)
  - **What:** examines if the system in question abides by the agreed design heuristic
  - **Why:** to see if the system can comfortably be used based on prior experience in similar systems

- **User Testing**
  - **Who:** end-user
  - **What:** uses the system for its intended purpose, as if he/she would be in a realistic situation at home/work/etc.
  - **Why:** to give direct input on how real users use the system